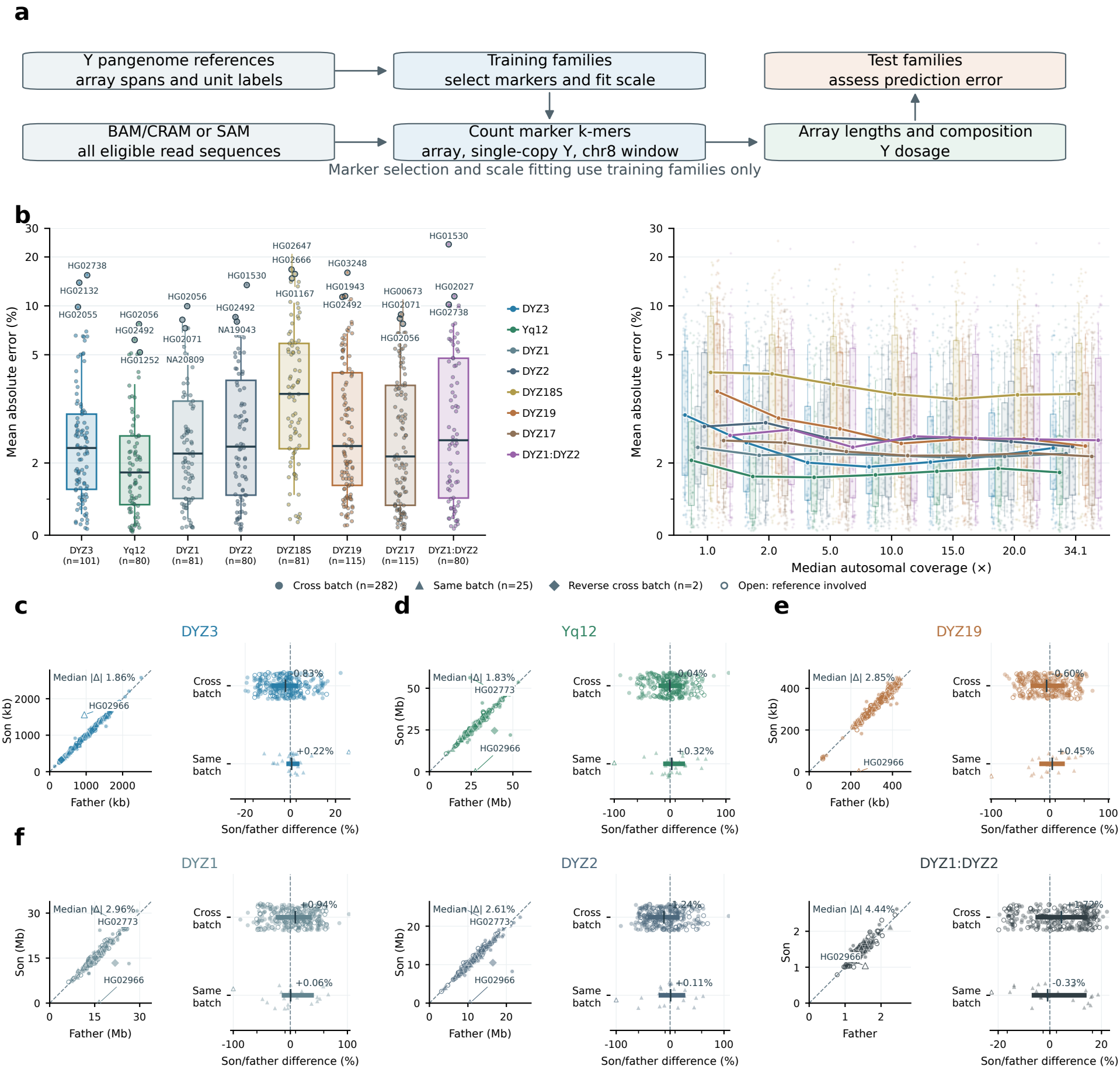

a
Y pangenome references
array spans and unit labels
Training families
select markers and fit scale
Test families
assess prediction error
BAM/CRAM or SAM
all eligible read sequences
Count marker k-mers
array, single-copy Y, chr8 window
Array lengths and composition
Y dosage
Marker selection and scale fitting use training families only
b
Mean absolute error (%)
DYZ3 (n=101)
Yq12 (n=80)
DYZ1 (n=81)
DYZ2 (n=80)
DYZ18S (n=81)
DYZ19 (n=115)
DYZ17 (n=115)
DYZ1:DYZ2 (n=80)
DYZ3
Yq12
DYZ1
DYZ2
DYZ18S
DYZ19
DYZ17
DYZ1:DYZ2
Median autosomal coverage (×)
Cross batch (n=282)
Same batch (n=25)
Reverse cross batch (n=2)
Open: reference involved
c
DYZ3
Median |Δ| 1.86%
HG02966
Son (kb)
Father (kb)
Cross batch
Same batch
-0.83%
+0.22%
Son/father difference (%)
d
Yq12
Median |Δ| 1.83%
HG02773
HG02966
Son (Mb)
Father (Mb)
-0.04%
+0.32%
e
DYZ19
Median |Δ| 2.85%
-0.60%
+0.45%
f
DYZ1
Median |Δ| 2.96%
+0.94%
+0.06%
DYZ2
Median |Δ| 2.61%
-1.24%
+0.11%
DYZ1:DYZ2
Median |Δ| 4.44%
Son
Father
+1.72%
-0.33%